

ReactiveGWM: Steering NPC in Reactive Game World Models

<https://inv-wzq.github.io/ReactiveGWM/>

Zeqing Wang^{12♣} Danze Chen^{12♣} Zhaohu Xing⁴ Zizhao Tong^{15♣} Yinhan Zhang^{16♣}
Xingyi Yang^{*3} Yeying Jin^{*12◇}

¹ Tencent ²National University of Singapore ³The Hong Kong Polytechnic University

⁴The Hong Kong University of Science and Technology (Guangzhou)

⁵University of Chinese Academy of Sciences

⁶The Hong Kong University of Science and Technology

zeqing.wang@u.nus.edu xingyi.yang@polyu.edu.hk jinyeying@u.nus.edu

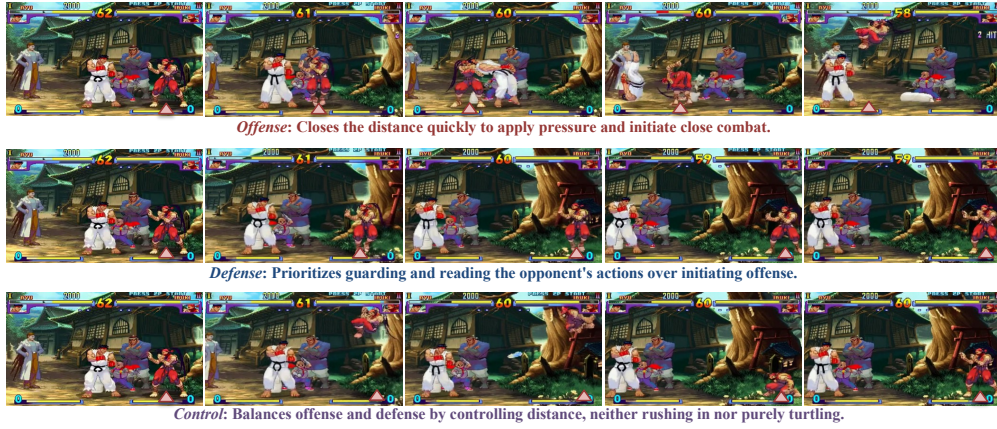


Figure 1: Visualization of the steerable NPC executing distinct strategies in *Street Fighter Alpha 3* (SF3) game. The NPC is denoted by $\blacktriangle$ triangle.

Abstract

Current game world models simulate environments from a subjective, player-centric perspective. However, by treating the Non-Player Character (NPC) merely as background pixels, these models cannot capture interactions between the player and NPC. In that sense, they act as passive video renderers rather than real simulation engines, lacking the physical understanding needed to model action-induced NPC reactivities. We introduce **ReactiveGWM**, a reactive game world model that synthesizes dynamic interactions between the player and NPC. Instead of entangling all interaction dynamics, ReactiveGWM explicitly decouples player controls from NPC behaviors. Player actions are injected into the diffusion backbone via a lightweight additive bias, while high-level NPC responses (e.g., Offense, Control, Defense) are grounded through cross-attention modules. Crucially, these modules learn a *game-agnostic representation* of interactive logic. This enables *zero-shot strategy transfer*: our learned modules can be plugged directly into off-the-shelf, unannotated world models of different games. This instantly unlocks steerable NPC interactions without any domain-specific retraining. Evaluated on two *Street*

♣This work was completed during a research internship at Tencent, supervised by Yeying Jin.

◇Project leader.

*Corresponding Author.

Fighter games, ReactiveGWM maintains fine-grain player controllability while achieving robust, prompt-aligned NPC strategy adherence, paving the way for scalable, strategy-rich interaction with the NPC.

1 Introduction

Recent advancements in world models [3, 12] have established a new paradigm for simulating complex environments. By capturing the underlying dynamics from vast amounts of offline gameplay videos, this paradigm naturally extends to the development of game world models [33, 29], unlocking unprecedented possibilities for interactive game generation.

However, most existing game world models fail to explicitly model the *Non-Player Character (NPC)*. Instead, they simulate environments from a player-centric perspective [5, 33, 17]. These models use player-centric prompts to generate non-player elements as part of the interactive background. Such a design implicitly assumes a deterministic relation between the player and the background. As a result, NPCs are often reduced to background pixels rather than modeled as dynamic and autonomous agents since their behaviors are tightly tied to fixed action sequences specified in the prompt. This makes most existing game world models closer to passive video renderers than to real game simulation engines. In actual games, NPCs follow high-level strategies to achieve dynamic and autonomous engagement. Ignoring this aspect limits gameplay to a largely solitary experience and prevents meaningful competitive interaction between the player and NPCs.

To overcome this limitation, we introduce a novel reactive game world model, termed **ReactiveGWM**. ReactiveGWM is explicitly designed to synthesize dynamic interactions between a player and an autonomous NPC. To achieve this, we construct novel datasets that decouple NPC autonomy from player control. In these datasets, in addition to gameplay videos and player action labels, each sample includes a structured prompt that only guides the NPC. Instead of entangling all interaction dynamics in standard player-centric prompts, our prompts specify the NPC with explicit strategic guidance and both active and passive behaviors, enabling autonomous strategy execution.

Given these structured, strategy-aligned datasets, we train ReactiveGWM to encode player control and NPC autonomy without entangling roles. Specifically, player actions are injected into the video diffusion backbone via a lightweight additive bias. Concurrently, we ground high-level NPC strategies in the cross-attention modules. Training on these data allows the cross-attention modules to learn a *player-agnostic representation* of interaction logic. Crucially, to enforce strategic autonomy rather than player-centric guidance, these modules are driven entirely by pure NPC behavioral (e.g., Offense, Control, Defense). This disentangles the NPC’s tactical intent from shallow descriptive prompts. Meanwhile, the game-specific physical and visual dynamics are still modeled by the original self-attention and feed-forward layers. By separating NPC behavioral logic from these dynamics, the learned behavior modules also form a *game-agnostic representation*. This representation can be transferred across different games in a plug-and-play manner.

Evaluated on two distinct *Street Fighter* games, our experiments show that ReactiveGWM maintains fine-grained player controllability while enabling autonomous, strategy-aligned NPC behavior. The results demonstrate realistic and dynamic interactions between the player and the NPC. More importantly, ReactiveGWM exhibits strong *zero-shot strategy transferability*. The learned NPC autonomy modules can be directly plugged into off-the-shelf vanilla world models of different games without additional annotation. This enables steerable NPC interactions without domain-specific strategy retraining, while preserving the native dynamics of the target game.

In summary, our contributions are summarized as follows: (1) We propose ReactiveGWM, breaking the limitations of player-centric modeling by simultaneously supporting fine-grained player control and strategy-driven NPC autonomy. (2) We construct new strategy-aligned datasets to explicitly distinguish tactical intent of the NPC from pixel-level rendering. With decoupled injection for player and NPC control, ReactiveGWM achieves both fine-grained player controllability and strategy-aligned NPC behavior. (3) We demonstrate that our specialized modules learn a *game-agnostic* interactive logic. These modules can be seamlessly transferred to off-the-shelf, unannotated target games, paving the way for highly scalable, strategy-rich game generation.

2 Related works

2.1 Controllable video generation

With the rapid advancement of video diffusion models [35, 20, 25, 24, 4, 44, 22, 37], visual content generation has achieved unprecedented fidelity. While detailed text prompts enable customized generation, they inherently lack fine-grained control, frequently resulting in spatiotemporal ambiguities. To achieve rigorous spatial and temporal alignment, controllable video generation frameworks incorporate auxiliary conditions, such as motion priors and trajectory inputs [36, 8, 45, 41, 48], camera trajectories [39, 43, 16], and structural guidance for consistent character animation [30, 10, 18, 19, 42, 49].

Beyond such localized controllability, a more ambitious line of work seeks to actively simulate causal physical mechanics, with the generative paradigm naturally evolving toward world models. By predicting future states and environmental transitions conditioned on current observations and external interventions [5, 23, 3], world models equip agents with a predictive “mental model” of the physical world. This capability is foundational for downstream decision-making, facilitating strategic planning and “learning in imagination” [12, 11, 13, 15]. Consequently, it has been shown to enable sample-efficient policy optimization in reinforcement learning and robotics [14, 28, 40], mitigating the cost of exhaustive interactions with the actual environment.

2.2 Game world models

Game world models aim to construct simulations of game environments, predicting future visual frames conditioned on player inputs. Pioneering works like GameNGen [34] demonstrated that diffusion models can serve as real-time neural engines for DOOM, while DIAMOND [1] established that the visual fidelity of diffusion world models significantly impacts downstream policy learning. Subsequent efforts, including Matrix-Game 2.0/3.0 [17, 29], LingBot-World [33], GameFactory [46], and Oasis [9], have pushed the boundaries toward streaming, long-horizon, and open-domain generation.

However, the conditioning vocabulary in the majority of these models remains uniformly restricted to the primary player’s action stream. Consequently, the Non-Player Character (NPC) are fundamentally absorbed into the background environmental dynamics without any explicit channel for high-level tactical intent or strategy following. Under the world-model paradigm, NPC behavior thus manifests merely as a passive byproduct of the training distribution, severely compromising NPC autonomy and neglecting a core interactive element of complex gameplay.

3 Method

3.1 Preliminaries

Most existing game world models [17, 46, 33] target the game-environment simulation from a player-centric view. Given an initial observation frame x_0 and a sequence of player actions $\mathbf{a}_T = \{a_0, a_1, \dots, a_{T-1}\}$, a vanilla world model $\mathcal{F}_{\text{vanilla}}$ predicts the future frames $\mathbf{x}_{1:T} = \{x_1, x_2, \dots, x_T\}$. The generation is conditioned on a player-centric prompt $\mathcal{P}_{\text{vanilla}}$:

$$\mathbf{x}_{1:T} = \mathcal{F}_{\text{vanilla}}(x_0, \mathbf{a}_T, \mathcal{P}_{\text{vanilla}}). \quad (1)$$

Here, $\mathcal{P}_{\text{vanilla}}$ typically describes the full scene, including background entities and player-related events, as shown in Figure 2. This formulation entangles the dynamics of the player and NPC within a single descriptive prompt. As a result, NPCs are not modeled as independent agents. They are instead treated as part of the visual background, with their behaviors implicitly tied to the vanilla prompt. This makes existing models closer to passive video renderers than to game simulation engines.

To enable steerable NPC behavior, the key is to decouple NPC behavior from $\mathcal{P}_{\text{vanilla}}$. We replace $\mathcal{P}_{\text{vanilla}}$ with an NPC-specific strategy prompt $\mathcal{P}_{\text{NPC}}$. This prompt does not describe all scene events. Instead, it provides high-level guidance for the NPC, such as tactical intent and behavior mode. Under this design, the model must account for two complementary factors: fine-grained player control and strategy-driven NPC autonomy. Hence, the generation process is written as

$$\mathbf{x}_{1:T} = \mathcal{F}(x_0, \mathbf{a}_T, \mathcal{P}_{\text{NPC}}). \quad (2)$$

Here, $\mathcal{P}_{\text{NPC}}$ acts as a high-level strategic instruction that governs the NPC’s decision-making process and interaction patterns, as shown in Figure 2. By incorporating this strategic prompt, the generated

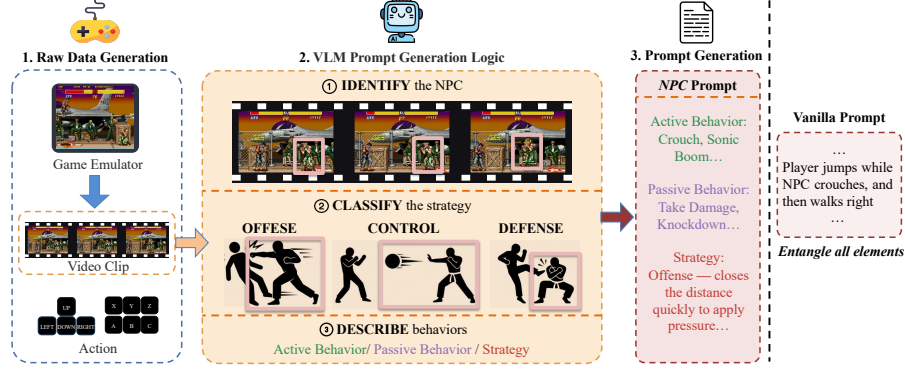


Figure 2: Overview of the data construction and strategy annotation pipeline. Each sample consists of a triplet: a video clip, player actions, and an NPC prompt. Unlike a vanilla prompt that entangles the dynamics between the player and NPC, the NPC prompt provides high-level strategy guidance.

video sequence $\mathbf{x}_{1:T}$ not only reflects the direct consequences of player actions $\mathbf{a}_T$ but also exhibits autonomous NPC behaviors that consistently follow the provided strategy.

In Section 3.2, we describe how to construct training triplets $(\mathbf{x}_{0:T}, \mathbf{a}_T, \mathcal{P}_{\text{NPC}})$. In Section 3.4, we present the training procedure and demonstrate the model’s ability to transfer and generalize autonomous NPC behaviors.

3.2 Data construction

We select *Street Fighter II: Champion Edition* (SF2) [6] and *Street Fighter Alpha 3* (SF3) [7] as our primary testbeds to construct our datasets. The whole data construction pipeline is shown in Figure 2.

Gameplay Recording. We employ the `stable-retro` [26] framework to programmatically collect gameplay episodes. A random agent uniformly samples from 10 discrete action buttons (directional movements and attacks). Episodes run until a round-end knock-out and are segmented into 5-second clips (20 fps). Each clip yields two aligned streams: a video clip $\mathbf{x}_{0:T}$ at native resolution, and a frame-level action record $\mathbf{a}_T$ structured as binary button-press vectors.

NPC Strategy Annotation. To derive $\mathcal{P}_{\text{NPC}}$, a Vision-Language Model (Gemini [31]) analyzes each clip to produce structured behavioral annotations. These encompass *active behaviors* (e.g., punch, kick, projectiles), *passive behaviors* (e.g., blocking, hit-stun), and a *strategy category* drawn from three mutually exclusive classes: **Offense** (closing distance to dominate melee), **Control** (maintaining distance via projectiles), and **Defense** (reactive, crouching guard). The final $\mathcal{P}_{\text{NPC}}$ is formulated as:

$$\mathcal{P}_{\text{NPC}} = \{\text{Active}(\dots), \text{Passive}(\dots), \text{Strategy}(\text{category}, \text{description})\} \quad (3)$$

This yields a complete triplet $(\mathbf{x}_{0:T}, \mathbf{a}_T, \mathcal{P}_{\text{NPC}})$ per clip, as shown in Figure 2. Through this pipeline, we curate $\sim 10,000$ training triplets per game. Further details are provided in Appendix A.

3.3 Model architecture

To condition frame generation on discrete actions $a_t \in \{0, 1\}^K$ (where $K=10$ for both SF2 and SF3), we adopt a lightweight additive bias mechanism instead of introducing heavy adapters or cross-attention modules [17, 29].

Let T denote the number of input video frames and T_v the temporal compression ratio of the VAE, so that the latent temporal length is $f = T/T_v$. The raw button sequence $a_{1:T} \in \{0, 1\}^{T \times K}$ is aligned to the latent frame rate via adaptive max-pooling along the time axis: the T frames are partitioned into f contiguous, nearly-equal bins $B_i = [\lfloor iT/f \rfloor, \lceil (i+1)T/f \rceil)$ for $i = 0, \dots, f-1$, and

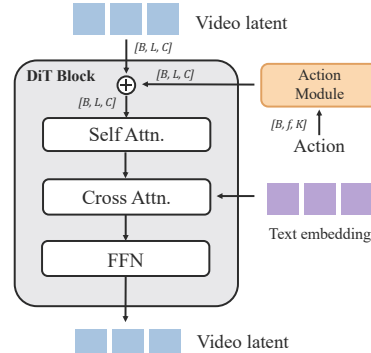


Figure 3: DiT block with action module.

each button channel takes the maximum within its bin:

$$\bar{a}_{i,k} = \max_{t \in B_i} a_{t,k}, \quad i \in [0, f), k \in [0, K).$$

The result $\bar{a} \in \{0, 1\}^{f \times K}$ forms a tensor of shape $[B, f, K]$, where B is the batch size.

To inject the action signal into the video backbone, we attach an independent, bias-free linear projection $E_\ell : \mathbb{R}^K \rightarrow \mathbb{R}^C$ to each DiT block ℓ , mapping the action representation to the hidden channel dimension C . The projected action embedding is then spatially broadcast across the $h \times w$ patch grid to match the flattened token sequence length $L = f \times h \times w$. This results in an action bias tensor of shape $[B, L, C]$, which is directly added to the video latent $x^{(\ell)} \in \mathbb{R}^{B \times L \times C}$ in the residual stream before the self-attention layer:

$$x^{(\ell)} \leftarrow x^{(\ell)} + E_\ell(\bar{a}) \otimes \mathbf{1}_{h \times w}. \quad (4)$$

3.4 ReactiveGWM

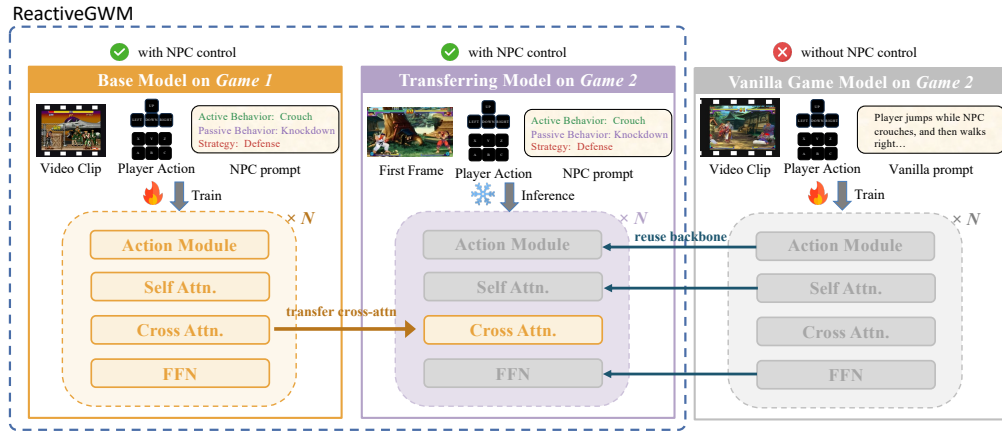


Figure 4: Overview of ReactiveGWM training and training-free transfer to the different game.

Based on the structured, strategy-aligned dataset described in Section 3.2 and the model architecture introduced in Section 3.3, we train ReactiveGWM to simulate game worlds with autonomous NPCs. As shown in Figure 4, our framework supports fully supervised training on a source game, denoted as $\text{ReactiveGWM}_{\text{base}}$, and further enables efficient training-free strategy transfer to different games, denoted as $\text{ReactiveGWM}_{\text{transfer}}$.

Model Training. For the source environment (denoted as Game 1), we perform full-parameter fine-tuning on the entire model architecture using the fully annotated strategy dataset to obtain $\text{ReactiveGWM}_{\text{base}}$. Specifically, all sub-modules within the DiT blocks (Figure 3)—including the Action Module, Self-Attention, Cross-Attention, and Feed-Forward Network (FFN)—are jointly optimized. Crucially, the Cross-Attention layers serve to ground the textual NPC strategy, $\mathcal{P}_{\text{NPC}}$, into the visual-temporal latent space, establishing a robust alignment between high-level linguistic tactics and low-level physical dynamics.

Autonomous NPC Transfer. Acquiring dense, frame-aligned strategy annotations for every new game is prohibitively expensive. To circumvent this scalability bottleneck, ReactiveGWM exhibits a powerful plug-and-play transfer capability. Suppose we have a $\mathcal{F}_{\text{vanilla}}$ pre-trained on a target environment (Game 2) using only standard $\mathcal{P}_{\text{vanilla}}$. To endow this vanilla model with steerable NPC capabilities, we can construct a $\text{ReactiveGWM}_{\text{transfer}}$ by composing modules from both models.

Specifically, we reuse the domain-specific backbone from the Game 2 vanilla model—retaining its pre-trained Action Module, Self-Attention layers, and FFN—to preserve the native physical and visual dynamics of Game 2. We then directly transfer and inject the learned Cross-Attention layers from the Game 1 NPC model into this backbone. Because the Cross-Attention modules encapsulate a generalized mapping for NPC control, this modular substitution enables zero-shot strategy conditioning in Game 2, entirely bypassing the need for new annotated strategy data. A detailed analysis of the factors underlying successful transfer is provided in Section 4.4.

4 Experiments

4.1 Setups

Dataset. Our strategy-aligned training dataset constructed via the pipeline in Section 3.2, comprises approximately 10k action-annotated video clips per game. Training resolutions are standardized to 480×608 for SF2 and 480×832 for SF3. To evaluate transferability, we additionally curate a *vanilla dataset* of equal scale (10k clips per game) utilizing standard descriptive prompts, which serves to train baseline game world models.

Model. We adopt the Wan2.2-TI2V-5B model [35] as the backbone video world model. Following Section 3.3, we augment the DiT architecture with the proposed action module to inject discrete player actions. Two models are trained under different supervision: **Vanilla Model**, trained on the vanilla dataset using standard prompts $\mathcal{P}_{\text{vanilla}}$, and **ReactiveGWM_{base}**, trained on the customized strategy dataset using strategy prompts $\mathcal{P}_{\text{NPC}}$. **ReactiveGWM_{transfer}** is the transferred model by transferring a trained model to a vanilla model.

Evaluation Metrics. To evaluate granular player action controllability, NPC autonomy, and spatiotemporal visual fidelity, we propose a three-dimensional framework (details are in Appendix B):

- **Player Action Following:** Evaluates strict adherence to input action sequences using a 100-run test set (10 initial frames $\times$ 10 single-key actions, 41 frames each).
 - *Movement Accuracy (Move-Acc):* Quantifies movement via SAM2.1 [27] and Grounding DINO [21] tracking. Success is defined by spatial displacement thresholds within a normalized $[0, 1]$ coordinate space.
 - *Attack Accuracy (Att-Acc):* Assessed by ClipAttackNet (ResNet-18 with a 4-layer dilated TCN [2]), a custom 6-way classifier trained on $\sim 5k$ clips. It predicts attack categories frame-wise with a 0.7 confidence threshold.
- **NPC Strategy Following:** We construct a benchmark using a fixed evaluation set of 99 curated clips (33 clips per tactical category: Control, Defense, and Offense). A Vision-Language Model (VLM) referee ensemble, comprising Gemini [31] and Qwen3-VL-8B [32], evaluates the generated 101-frame video sequences to compute:
 - *Categorical Accuracy:* The 3-way top-1 match rate between VLM predictions and ground-truth strategies.
- **Visual Quality:** Evaluates long-term fidelity using the aforementioned 99 clips. We compare 101-frame generated videos against ground-truth game engine outputs:
 - *SSIM [38]:* Frame-averaged Structural Similarity Index Measure for structural distortions.
 - *LPIPS [47]:* Full-frame Learned Perceptual Image Patch Similarity (AlexNet backbone) for perceptual fidelity.

Baselines. We compare ReactiveGWM with the Matrix-Game-3.0 [29] and LingBot-World-Base (Act) [33] baselines. Notably, due to architectural differences in their action injection mechanisms, we restrict their evaluation strictly to NPC Strategy Following and Image Quality. Furthermore, because these baselines are not explicitly tailored for the SF2 and SF3 environments, they serve primarily as a broad reference. **Consequently, the core of our evaluation focuses on analyzing the Vanilla model and ReactiveGWM.**

4.2 Main results

As summarized in Table 1, we evaluate ReactiveGWM against baselines across the three proposed dimensions: Action Control, NPC Strategy Following, and Visual Quality. The results demonstrate that our approach successfully imbues the world model with high-level NPC autonomy with high visual fidelity and player controllability. A user study is provided in Appendix D.

Superior NPC Autonomy. ReactiveGWM substantially improves the expression of strategic NPC intent. Compared with the vanilla model, the VLM-judged instruction accuracy increases from $\sim 43\%$ to over 75% on SF2, and from $\sim 41\%$ to $\sim 79\%$ on SF3. These results show that the NPC strategy

Table 1: Quantitative comparison of game world models.

Method	Action Control (%)		NPC Instruction (%)		Visual Quality	
	Move-Acc $\uparrow$	Att-Acc $\uparrow$	Gemini $\uparrow$	Qwen $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
SF2						
Matrix-Game-3.0 [29]	-	-	3.0	24.2	0.084	0.755
LingBot-World-Base [33]	-	-	30.3	46.5	0.142	0.679
Vanilla	97.5	96.7	43.4	44.4	0.427	0.315
ReactiveGWM_{base}	95.0	93.3	75.8	76.8	0.428	0.319
ReactiveGWM_{transfer}	97.5	93.3	64.6	64.6	0.421	0.318
SF3						
Matrix-Game-3.0 [29]	-	-	32.5	32.3	0.117	0.685
LingBot-World-Base [33]	-	-	33.5	40.9	0.202	0.572
Vanilla	100.0	100.0	41.8	49.5	0.392	0.397
ReactiveGWM_{base}	100.0	100.0	79.8	78.8	0.394	0.391
ReactiveGWM_{transfer}	95.0	100.0	63.6	73.7	0.367	0.414

prompt $\mathcal{P}_{\text{NPC}}$ provides an explicit signal for tactical intent, moving NPC behavior beyond passive environmental dynamics.

Figures 8 and 1 further show that ReactiveGWM_{base} follows three distinct tactical directives. Under the ‘Offense’ strategy, the NPC actively approaches the player and engages in close combat. Under the ‘Defense’ strategy, the NPC keeps a safe distance and reacts evasively to the player’s actions. Under the ‘Control’ strategy, the NPC zones the player with ranged projectile attacks, such as Sonic Boom in *SF2* (e.g., the third and fifth frames in the bottom row of Figure 8) and airborne projectiles in *SF3* (e.g., the third frame in the bottom row of Figure 1). Visual comparisons with Matrix-Game-3.0 and LingBot-World-Base are provided in Appendix C.

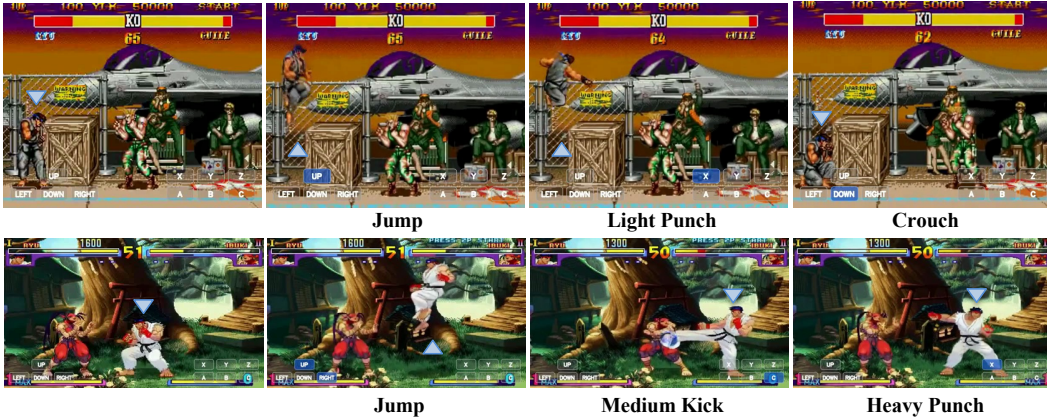


Figure 5: Action control in ReactiveGWM_{base}. The player-controlled character is denoted by a $\blacktriangle$ triangle. The specific action button mappings for each game are detailed in Appendix A.

Preserved Control and Fidelity. Crucially, empowering NPC autonomy does not compromise core mechanics. For single-action testing, ReactiveGWM maintains near-perfect Action Control (e.g., 100.0% Move-Acc and Att-Acc in *SF3*) and visual quality (SSIM/LPIPS), remaining strictly on par with the vanilla baseline. For sequence actions, as qualitatively demonstrated in Figure 5, the model precisely adheres to diverse, fine-grained player commands. The player-controlled character (indicated by the blue triangle) flawlessly executes spatial movements (e.g., Jump, Crouch) and distinct combat actions (e.g., Light Punch, Medium Kick, Heavy Punch) across two game domains. This visual evidence, coupled with our quantitative results, confirms that our architecture effectively disentangles explicit player interventions from autonomous NPC strategies without degrading rendering fidelity.

Transferability. $\text{ReactiveGWM}_{\text{transfer}}$ fully retains the high action controllability (e.g., 97.5% Move-Acc in SF2) and visual quality of the base models while delivering competitive NPC Strategy Following (up to 73.7% in SF3). This demonstrates that complex strategy compliance can be efficiently transferred to a vanilla model without exhaustive full-parameter retraining.

Offense: Maintains constant aggression to overwhelm the opponent and force defensive



Figure 6: Comparison between the vanilla model and ReactiveGWM under the same strategy. The NPC is indicated by the $\blacktriangle$ triangle.

Figure 6 compares the vanilla model, $\text{ReactiveGWM}_{\text{base}}$, and $\text{ReactiveGWM}_{\text{transfer}}$ under the same strategy. The vanilla model fails to follow the ‘Offense’ directive, whereas both variants of ReactiveGWM produce strategy-consistent NPC behavior.

4.3 Prompt analysis

While our main results demonstrate the overall strategy-following capabilities of our model, this section investigates the specific impact of *active behaviors* on the autonomous NPC. Essentially, *active behaviors* supply the executable actions required to successfully realize the defined high-level strategies. To verify the effectiveness of this component, we present qualitative visualizations of representative scenarios.

As specifically illustrated in Figure 7, we evaluate the model’s capacity to guide NPC followed by *active behaviors*. In the top row, guided by the prompt “Standing Punch + Throw”, the NPC accurately performs a punch followed seamlessly by a close-range grapple. The middle row (“Jumping Attack + Standing Punch”) successfully initiates an aerial assault and immediately follows up with a grounded punch upon landing. Finally, the bottom row (“Standing Kick + Crouching Kick”) showcases fine-grained postural control, where the NPC flawlessly transitions from a standing kick to a crouching low kick.

4.4 Transferring analysis

To systematically understand why transferring the Cross-Attention module achieves both high-fidelity visual preservation and effective non-player character (NPC) strategy control, we conduct quantitative analysis for $\text{ReactiveGWM}_{\text{Vanilla}}$ and $\text{ReactiveGWM}_{\text{Transfer}}$ in SF2. Both models share the same DiT backbone. Under identical seed, init frame, prompt template, we run inference for each model on three strategies (Offense, Defense, Control), yielding $2 \times 3 = 6$ forward passes (101 frames at 480×608 , 30 diffusion steps each).



Figure 7: Execution of *active behaviors*. The NPC (indicated by the $\blacktriangle$ triangle) is successfully guided to perform specific defined actions.

Visual Preservation. Table 1 indicates that $\text{ReactiveGWM}_{\text{Transfer}}$ retains the original visual subjects and scenes. This phenomenon is primarily attributed to the structurally low-bandwidth nature of the Cross-Attention layer and the high directional compatibility of its output. Within each transformer block ℓ , the visual token residual x_ℓ is updated via three pathways: Self-Attention (SA), Cross-Attention (CA), and Feed-Forward Networks (FFN). We quantify the relative energy share of the cross-attention injection as:

$$\rho_\ell^{\text{cross}} = \frac{\|CA_\ell\|^2}{\|SA_\ell\|^2 + \|CA_\ell\|^2 + \|FFN_\ell\|^2} \quad (5)$$

Our measurements reveal that $\overline{\rho^{\text{cross}}}$ is merely 0.71% for $\text{ReactiveGWM}_{\text{Transfer}}$, nearly identical to the Vanilla model (0.70%). This indicates that structurally, the Cross-Attention acts as a low-bandwidth channel. The remaining $\sim 99.3\%$ of the energy, which dictates the main visual components, is left entirely undisturbed.

NPC Control. Despite the low channel bandwidth, $\text{ReactiveGWM}_{\text{Transfer}}$ can still steer NPC behaviors (e.g., Offense, Defense, Control). We observe that the transferred module introduces a new signal direction that enables controllable NPC behavior.

We define the directional difference as $\Delta_\ell := CA_\ell^T - CA_\ell^V$. The token-averaged cosine similarity $\cos(CA_\ell^V, CA_\ell^T)$ drops to 0.55, indicating the emergence of a substantially different signal direction. Accumulated over 30 blocks and 30 diffusion steps, this directional signal is sufficient to steer NPC trajectories without interfering with the dominant visual dynamics.

5 Conclusion

We presented ReactiveGWM, a reactive game world model for simulating autonomous NPC behavior. Unlike prior player-centric world models, ReactiveGWM separates NPC autonomy from player control. Player actions are injected into the diffusion backbone through a lightweight additive bias, while high-level NPC strategies are grounded through cross-attention modules. This design allows the model to preserve fine-grained player controllability while generating strategy-aligned NPC responses. To support this formulation, we constructed strategy-aligned datasets that pair gameplay videos and player actions with NPC-specific prompts. These prompts provide high-level tactical guidance, such as Offense, Control, and Defense, instead of describing all scene dynamics in a

single vanilla prompt. Experiments on two *Street Fighter* games show that ReactiveGWM produces more autonomous and prompt-consistent NPC behavior than vanilla game world models. Moreover, the learned NPC behavior modules can be transferred to off-the-shelf world models of different games without additional annotation or retraining. This demonstrates that the modules capture a game-agnostic representation of interaction logic. Overall, by enabling steerable NPC autonomy and zero-shot strategy transfer, ReactiveGWM provides a step toward scalable, strategy-rich game generation.

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A Data construction

This appendix details the data construction pipeline introduced in Section 3.2.

A.1 Action space and recording setup

Both SF2 and SF3 are emulated through `stable-retro` with a fixed Player vs. NPC matchup. The emulator exposes a 12-bit button vector; of these we use **10 physical buttons**—4 directional (UP, DOWN, LEFT, RIGHT) and 6 attack buttons (A, B, C, X, Y, Z). On top of these raw buttons we define **13 discrete behaviors** (IDs 0–12) that cover the full strategic range of both games (Table 3).

The two titles use different physical pads, so the six attack keys A/B/C/X/Y/Z map to different punch/kick strengths in each game. We give the per-game button-to-semantic mapping in Table 2, and the final behavior-to-button table in Table 3; the behavior IDs themselves are shared.

Table 2: Per-game mapping from the six attack buttons exposed by `stable-retro` to punch/kick semantics. Light/Medium/Heavy are abbreviated LP/MP/HP for punches and LK/MK/HK for kicks.

Button	A	B	C	X	Y	Z
SF2	LK	MK	HK	LP	MP	HP
SF3	LP	LK	MK	HP	MP	HK

Table 3: The 13 discrete behaviors and the physical buttons pressed in each game. Directional keys are shared across the two titles; only the attack buttons differ (Table 2). Behaviors 11 and 12 are forward/back jumps and combine UP with a horizontal key.

ID	Behavior	Semantic	SF2 buttons	SF3 buttons
0	No-op	idle	$\emptyset$	$\emptyset$
1	Walk Left	move left	LEFT	LEFT
2	Walk Right	move right	RIGHT	RIGHT
3	Jump	vertical jump	UP	UP
4	Crouch	crouch	DOWN	DOWN
5	Light Punch	LP	X	A
6	Medium Punch	MP	Y	Y
7	Heavy Punch	HP	Z	X
8	Light Kick	LK	A	B
9	Medium Kick	MK	B	C
10	Heavy Kick	HK	C	Z
11	Jump Right	forward jump	UP+RIGHT	UP+RIGHT
12	Jump Left	back jump	UP+LEFT	UP+LEFT

At the frame level each behavior is serialized with an EDGE/HOLD scheme: directional keys (LEFT, RIGHT, DOWN) are *held* for the entire 10-frame decision block, while attack and UP keys are emitted as a single-frame *edge* press at the start of the block. One decision block therefore corresponds to 10 video frames.

A.2 Episode recording and clip segmentation

A random agent samples behaviors uniformly from $\{0, \dots, 12\}$ until a round-end knockout. To exclude pre-round, we discard the first 5 s of each episode. The remaining footage is chunked into 5-second windows (100 frames at 20 FPS). Each clip yields a native-resolution `video.mp4` and a frame-level binary action table `actions.parquet` (`aT`) that records the 12-bit button vector per frame.

A.3 Two-stage NPC strategy annotation

High-level strategy is contextual and cannot be read from RAM. We therefore use a two-stage pipeline that separates *factual observation* from *categorical inference*, so that any residual VLM hallucination can only corrupt the facts, not the final label.

Stage 1 — Factual observation by a VLM. Gemini watches each 5s clip and answers 12 short, factual questions about the NPC (e.g., Guile in SF2), listed in Table 4. The prompt explicitly forbids the VLM from naming a strategy; it may only (i) report observable facts with a closed value set, and (ii) tag the NPC’s own moves from a fixed vocabulary, split into *active* (self-initiated attacks/motion) and *passive* (blocks/damage reactions). The 12 observations break down into six per-move facts (Q1–Q6, one per attack type), three engagement facts (Q7–Q9), and three aggregate facts (Q10–Q12).

Table 4: Stage 1 factual observations about the NPC. The VLM is prompted to answer each with a closed-set value; it never outputs a strategy label. We use Guile (the NPC in SF2) as an example.

Question	Values	Purpose
guile_does_punch	yes / no	melee detection
guile_does_kick	yes / no	melee detection
guile_does_jumping_attack	yes / no	melee detection
guile_does_throw	yes / no	melee detection
guile_does_flash_kick	yes / no	special-move detection
guile_sonic_boom_count	0 / 1 / 2+	projectile detection
guile_engagement_range	close / mid / far	spatial context
guile_advances	yes / no / unclear	positional intent
guile_takes_damage	yes / no	exchange outcome
guile_close_range_pressure	yes / no	sustained aggression
guile_crouches_guard	yes / no	defensive posture
who_attacks_more	ryu / guile / both / neither	initiative

Stage 2 — Deterministic classification. A rule engine maps the Stage 1 facts (and the active/passive tag lists) to one of three mutually exclusive strategies; clips that match no rule are dropped. Let $\text{melee} = [\text{punch}, \text{kick}, \text{jumping_attack}, \text{throw}, \text{flash_kick}]$ and $\text{has_melee} = \exists q \in \text{melee} : q = \text{yes}$. The rules are:

- **Offense:** $\text{range} = \text{close} \wedge \text{advances} = \text{yes} \wedge \text{has_melee} \wedge \text{sonic_boom} = 0 \wedge \text{pressure} = \text{yes} \wedge \text{who} = \text{guile}$.
- **Control:** $\text{range} \in \{\text{mid}, \text{far}\} \wedge \text{advances} = \text{no} \wedge \text{sonic_boom} \geq 1 \wedge \neg \text{has_melee} \wedge \text{takes_damage} = \text{no}$.
- **Defense:** $\neg \text{has_melee} \wedge \text{sonic_boom} = 0 \wedge \text{crouches_guard} = \text{yes} \wedge \text{who} \in \{\text{ryu}, \text{neither}\} \wedge \text{active} \subseteq \{\text{Crouch}, \text{Walk L}, \text{Walk R}\} \wedge \text{passive} \cap \mathcal{D} \neq \emptyset$, where $\mathcal{D}$ is the set of defensive passive tags (*Standing/Crouching Block, Take Damage, Knockback, Knockdown, Wake Up, Stun, Evade*).

The three rules are mutually exclusive by construction, so every surviving clip is assigned a unique label purely from unambiguous facts.

A.4 Prompt assembly

The final NPC guidance is assembled from Stage 1 tags and the Stage 2 label:

$$\mathcal{P}_{\text{NPC}} = \{ \text{Active}(b_1:d_1; \dots), \text{Passive}(b'_1:d'_1; \dots), \text{Strategy}(c:\delta_c) \}, \quad (6)$$

where b_i (resp. b'_i) are the active (resp. passive) vocabulary tags with descriptions d_i , $c \in \{\text{Offense}, \text{Control}, \text{Defense}\}$ is the Stage 2 label, and δ_c is a natural-language paraphrase deterministically drawn from a per-category paraphrase pool via $\text{MD5}(\text{video_path}) \bmod |\text{pool}_c|$. The hash selection guarantees bit-wise reproducibility while exposing the model to varied surface forms of the same strategy.

B Benchmark details

This section describes the evaluation protocol used in the main paper. We evaluate the models from three aspects: **Player Action Following**, **NPC Strategy Following**, and **Visual Quality**. All compared methods use the same frozen data splits, rollout settings, and preprocessing pipeline. Since SF2 and SF3 follow the same evaluation procedure, we only present the example of SF2.

B.1 Player action following

Goal. This dimension evaluates whether generated videos faithfully execute commanded player actions.

Evaluation set. We use a fixed 100-run benchmark built from 10 initial frames $\times$ 10 single-key actions ($\{\text{LEFT, RIGHT, UP, DOWN, Y, X, Z, A, B, C}\}$). Each action sequence is sparsely encoded with repeated key presses, and each rollout contains 41 generated frames.

Segmentation and trajectory extraction. For each generated clip, we run a player-character segmentation pipeline with SAM2.1 and Grounding DINO, then extract per-frame geometric trajectories from masks. For each character and frame, we compute:

$$(x, y, w, h, \text{area}, \text{aspect}),$$

where x, y are bbox center coordinates, w, h are bbox width/height, all normalized to $[0, 1]$, area is mask area ratio, and $\text{aspect} = h/w$.

Movement Accuracy (Move-Acc). Movement clips are scored by thresholded displacement rules on normalized coordinates. Let (x_0, y_0, h_0) be initial values from valid frames, and (x_T, y_T) be end-frame values. A 5-frame rolling median is applied before scoring for robustness to occasional mask jitter.

$$\text{LEFT: } x_T - x_0 \leq -0.025, \quad (7)$$

$$\text{RIGHT: } x_T - x_0 \geq +0.025, \quad (8)$$

$$\text{UP: } \min_t (y_t) - y_0 \leq -0.030 \quad (\text{peak-only}), \quad (9)$$

$$\text{DOWN: } (h_{\text{mid}} \leq 0.85 h_0) \vee (y_{\text{mid}} - y_0 \geq 0.010). \quad (10)$$

The UP criterion is peak-only because sparse repeated UP inputs may keep the character airborne near clip end. Move-Acc is the mean over the four movement keys.

Attack Accuracy (Att-Acc). We use ClipAttackNet, a 6-way attack classifier (ResNet-18 backbone + 4-layer dilated TCN head) trained on $\sim 5k$ labeled clips. Training uses a 3-stage fine-tuning schedule: (1) head-only training, (2) unfreeze layer4 + head, (3) unfreeze layer3/layer4 + head, with BCE-with-logits loss masked by valid frames. The validation checkpoint is selected by mean clip IoU at threshold 0.7.

At inference, each frame outputs 6-way probabilities. Frames with $\max p_k > 0.7$ are considered attack-active; clip prediction is the key of the most confident active frame. If no frame is attack-active, prediction is “noop”. Att-Acc is clip-level top-1 accuracy over attack keys.

B.2 NPC strategy following

Goal. This dimension evaluates whether generated NPC behavior follows high-level tactical intent.

Evaluation set. We evaluate on a frozen curated 99-clip subset with three categories: *Control*, *Defense*, and *Offense*. Each sample is evaluated as a 101-frame generated video.

VLM referees. Predictions are produced by two VLM referees (Gemini [31] and Qwen3-VL-8B [32]).

Categorical Accuracy: 3-way top-1 accuracy between predicted and ground-truth strategy categories on valid samples.

VLM Referee Prompts. For reproducibility, we include the exact decision-oriented prompts used by our two referee models. Both prompts require strict JSON-only output and the same output schema.

Prompt A (Gemini).

You are evaluating a Street Fighter II gameplay video. There are exactly two characters on screen:

PLAYER (Ryu):

- White karate gi (top + pants), red headband, brown belt, red gloves.
- Black hair, barefoot.

NPC (Guile) - the character you must analyze:

- Green/olive military tank top, green camouflage pants, red boots.
- Tall and muscular, distinctive blonde flat-top (high-and-tight) hair.

Analyze the NPC's behavior across the entire video and decide which strategy category fits best.

"Control" - uses ZONING tools to manage distance and space.

Trigger if ANY of the following:

- (a) Guile launches a Sonic Boom.
- (b) Guile holds an EXTENDED CROUCH at medium-to-far distance (NOT close-range melee or cornered), maintaining spacing.
- (c) Guile alternates brief forward/backward steps at medium range (spacing dance) without committing to close combat.

"Offense" - actively pressures the player:

- (i) sustained forward movement toward the player across multiple frames, OR
- (ii) TWO OR MORE distinct close-range attacks.

"Defense" - primarily passive/reactive:

standing, blocking, retreating, jumping back, close-range turtling, or a SINGLE reactive counter attack.

KEY DISTINCTION - Control vs Defense:

- Crouching AT A DISTANCE while keeping spacing -> Control.
- Crouching AT CLOSE RANGE / cornered / blocking -> Defense.

DECISION ORDER (first match wins):

- 1) Sonic Boom? -> Control
- 2) Extended distance crouch/zoning posture? -> Control
- 3) Sustained forward movement OR >=2 close-range attacks? -> Offense
- 4) Otherwise -> Defense

EDGE CASES:

- Post-match KO/defeat/victory animation for most of clip -> npc_visible=false
- Rendering broken or NPC missing/unidentifiable -> npc_visible=false
- Ryu fireball does NOT count as Guile Sonic Boom

Output EXACTLY this JSON object:

```
{
  "npc_side": "left" or "right",
  "npc_visible": true or false,
  "category": "Control" | "Defense" | "Offense",
  "category_reason": short string (<=30 words),
  "scene_description": one or two sentences describing NPC actions
}
```

Return ONLY the JSON object, no other text.

Prompt B (Qwen3-VL-8B).

You are watching a 5-second Street Fighter II gameplay clip with two characters.

PLAYER (Ryu): white karate gi, red headband, black hair.

NPC (Guile): green tank top, green camo pants, red boots, blonde flat-top hair.

Classify Guile's behavior into ONE category. Apply rules top-down (first match wins):

Rule 1 (Control, highest priority):

If Guile fired a Sonic Boom -> Control.

Rule 2 (Offense):
 If Guile clearly moved toward Ryu OR performed any close-range attack
 (even one punch/kick/sweep/jump-in/anti-air) -> Offense.

Rule 3 (Control fallback):
 If Guile stays at distance (not cornered, not close-range) and
 holds crouch or alternates crouch/stand -> Control.

Rule 4 (Defense, default):
 Otherwise (cornered/close-range passive blocking, retreat-only, etc.) -> Defense.

EDGE CASES:
 - Mostly post-match animation / broken rendering / NPC missing -> npc_visible=false
 - Only count Guile’s actions, not Ryu’s.

Output EXACTLY this JSON:

```
{
  "npc_side": "left" or "right",
  "npc_visible": true or false,
  "category": "Control" | "Defense" | "Offense",
  "category_reason": short string (<=25 words),
  "scene_description": one or two sentences of Guile’s observed actions
}
```

Return ONLY the JSON object.

B.3 Visual quality

Goal. This dimension measures long-horizon structural and perceptual fidelity of generated videos.

Evaluation set and temporal alignment. We reuse the same frozen 99 clips from NPC Strategy Following. Generated clips are 101 frames, while reference videos are 100 frames; evaluation uses the aligned window of $\min(101, 100) = 100$ frames.

Preprocessing alignment. To avoid interpolation bias, reference frames are transformed with the same center-crop + bicubic resize path used by model inference preprocessing, and compared at the target resolution.

Metrics.

- **SSIM:** frame-averaged structural similarity (higher is better).
- **LPIPS:** full-frame LPIPS with AlexNet backbone, averaged over frames (lower is better).

C Visualization

In addition to the ReactiveGWM results presented in Section 4.2, we visualize LingBot-World and Matrix-Game-3.0 in the *SF2* game scenario, as shown in Figure 9. The results indicate that neither model is well-suited to *SF2*, since they are not designed for this type of game. Therefore, we include them only as reference baselines in Table 1, and focus our analysis on the comparison between the vanilla model and ReactiveGWM.

D User study

We further conduct a human study on *Street Fighter II* (SF2) and *Street Fighter III* (SF3) to validate the main results in Section 4.2. The study evaluates two dimensions: **Player Action Following** and **NPC Strategy Following**. We recruit 19 participants who are familiar with 2D fighting games. Each participant completes the full questionnaire for both games. We report the mean Likert scores together with the standard error of the mean (SEM).

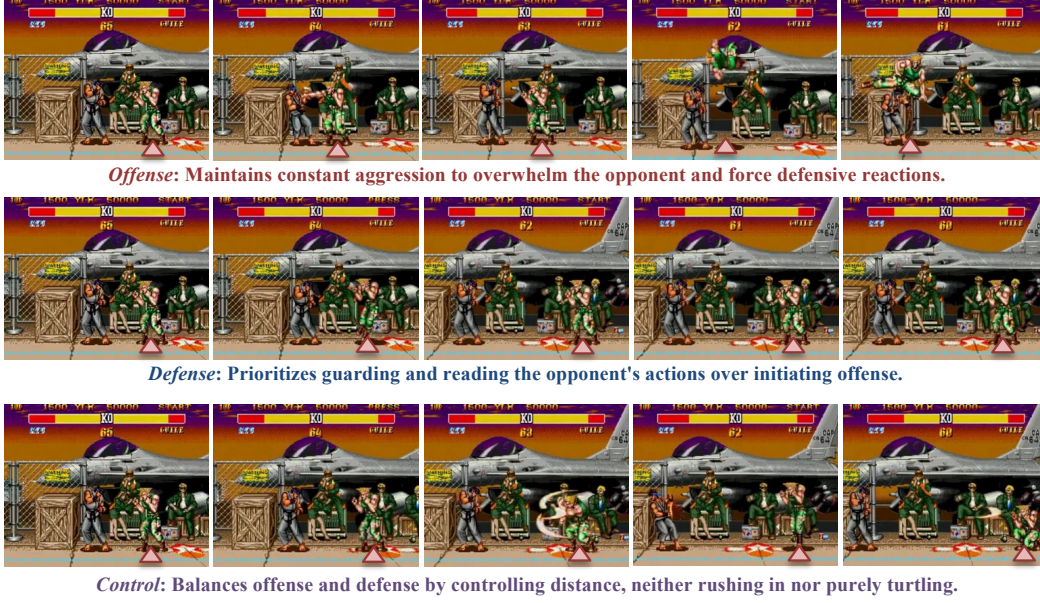


Figure 8: Visualization of the steerable NPC executing distinct strategies in *SF2* game. The NPC is denoted by the $\blacktriangle$ triangle.



Figure 9: Visualization of LingBot-World and Matrix-Game-3.0 on *SF2*.

D.1 Player action following

This part evaluates whether the on-screen *player character* faithfully follows the action inputs. Each participant watches a generated clip with its key-input overlay and assigns a 1–5 Likert score, where 1 indicates poor alignment, and 5 indicates full alignment.

As shown in Figure 10, all three generators obtain scores between 4.32 and 4.60 on *SF2* and *SF3*. In all conditions, the difference between any two models is within one SEM. These results suggest that player action control is reliable across all models, further supporting the effectiveness of our model architecture described in Section 3.3.

D.2 NPC strategy following

This part tests whether the high-level strategy prompt actually steers the generated NPC. Each clip is produced under one intended strategy chosen from *Control*, *Defense*, and *Offense*, and the participant selects the strategy that the NPC in the clip appears to be following. The classification accuracy

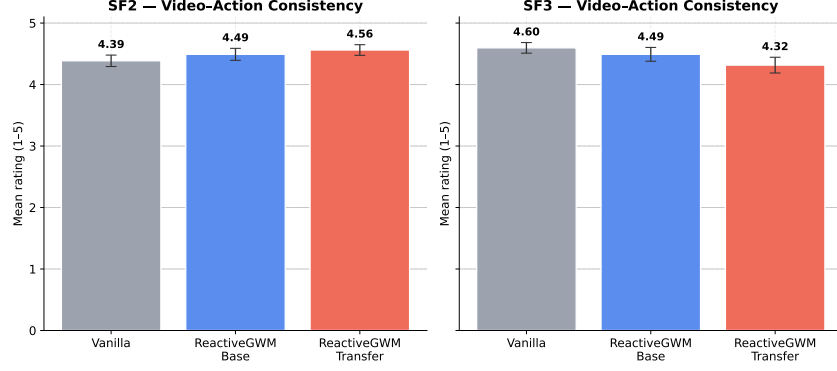


Figure 10: User Study Part 1 in *Player Action Following*. Mean participant scores ($\pm$ SEM) on a 1–5 Likert scale, where higher scores indicate better action following.

therefore directly measures how faithfully the generated NPC carries out the prompted strategy: a higher accuracy means the strategy condition successfully controls the NPC’s behaviour.

As shown in Figure 11, ReactiveGWM delivers a clear and consistent gain in this metric. On SF2, the overall accuracy reaches 86.0% for ReactiveGWM_{base} and 84.2% for ReactiveGWM_{transfer}, roughly double the 43.9% of the unconditioned Vanilla baseline. The gap widens further on SF3: Vanilla collapses to 17.5%, while ReactiveGWM_{base} climbs to 77.2% and ReactiveGWM_{transfer} to 61.4%, gains of 59.7 and 43.9 percentage points respectively.

The per-class breakdown tells the same story. Vanilla often fails on a specific strategy, scoring only 10.5% on *Offense* in SF2 and on both *Defense* and *Offense* in SF3, whereas ReactiveGWM_{base} stays above 63% on every class in both games. The only remaining weak spot of ReactiveGWM_{transfer} is the *Control* class on SF3, where its accuracy of 16% contrasts with 100% on *Offense*. This suggests that zoning behavior is the hardest axis to transfer across games. A likely reason is that *Control* often depends on game-specific ranged attacks. These attacks vary substantially across games in their animation, timing, trajectory, and spatial effect. As a result, the learned control strategy does not transfer as directly as more general behaviors such as offense and defense.

These results show that the strategy prompt provides an effective and human-perceivable handle on the generated NPC.

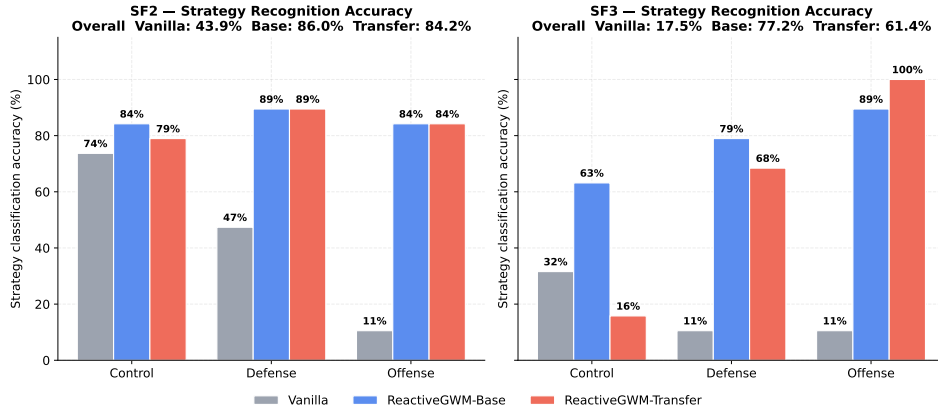


Figure 11: User Study Part 2 in *NPC Strategy Following*. Per-class and overall human strategy-classification accuracy.

E Limitations and future work

While ReactiveGWM shows robust strategy following and zero-shot transferability, it has two main limitations. First, our evaluation is limited to 2D fighting games. This genre offers a strong testbed for fine-grained action control and high-level tactics. However, extending the framework to other game categories, such as 2D FPS games or multi-agent strategy games, is needed to better assess the generality of the learned game-agnostic representations. Second, the diffusion-based backbone introduces high inference latency. This prevents a truly real-time interactive experience. To move from a reactive video renderer toward a fully playable game engine, future work should explore autoregressive video generation and model distillation. These directions may reduce inference latency while preserving visual quality and tactical fidelity.